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ASSIGNMENT 3

PROGRESS 3 – INTELLIGENT AGENT

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INTRODUCTION

The AI Task Automation System is designed to improve industrial efficiency by automating routine tasks and intelligently managing workflows. Many industries still depend on manual processes, which lead to slow task completion, human errors, and weak coordination across departments. To address these challenges, the system uses Artificial Intelligence to analyse operational conditions and decide the most suitable actions at any moment.

To further define the proposed AI Task Automation System, the PEAS model is used to formally describe the system as an intelligent agent. PEAS stands for Performance measure, Environment, Actuators, and Sensors, and it is commonly used in Artificial Intelligence to specify the behaviour and objectives of an AI system operating in a real-world environment.

By applying the PEAS framework, we will identify what the AI system is expected to achieve, the environment in which it operates, how it perceives changes, and how it takes actions. This representation ensures that the AI Task Automation System is goal-driven, measurable, and aligned with industrial requirements such as efficiency, accuracy, automation, and reliability.

In the context of industrial operations, the AI Task Automation System functions as an autonomous agent that continuously monitors tasks, machines, resources, and workflows, and then makes intelligent decisions such as task assignment, workflow triggering, predictive maintenance, and anomaly reporting. The intelligent agent provides a structured way to connect these capabilities with the previously defined Knowledge Representations and State Space Search, that will ensure consistency across the system design.

OVERVIEW OF PEAS

The AI Task Automation System acts as an intelligent agent that perceives its industrial environment through sensors, reasons using knowledge representation and state space search, and acts through actuators to achieve predefined goals. The agent's behaviour and effectiveness are formally described using the PEAS model.

P – Performance Measure

The performance measure defines how well the intelligent agent achieves its objectives in automating industrial and administrative workflows. The agent's performance is evaluated based on the following criteria:

1. Task Execution Efficiency

- Reduction in task completion time
- Automatic execution of dependent tasks without delay
- Increased number of completed tasks per unit time

2. Accuracy and Error Reduction

- Minimization of human errors in task assignment and data handling
- Correct execution of workflow dependencies
- Accurate anomaly detection

3. Operational Productivity

- Improved coordination between departments
- Reduced manual intervention
- Faster decision-making support

4. Resource Optimization

- Efficient use of manpower, materials, and energy
- Balanced workload allocation
- Reduction in resource wastage

5. Machine Reliability

- Early detection of machine faults
- Reduced downtime through predictive maintenance
- Stable machine operating conditions

6. User Satisfaction

- Ease of interaction with the system
- Reduced workload and operational stress
- Increased trust in AI-assisted decisions

E – Environment

The environment represents the setting in which the AI Task Automation System operates and interacts as an intelligent agent. It includes all internal and external elements that influence task execution, decision-making, and system performance. The environment of the AI Task Automation System consists of the following components:

1. Tasks and Workflows

- Production tasks, document approvals, inventory updates, and maintenance activities
- Task priorities, deadlines, and dependencies that define workflow order

2. Human Resources

- Factory operators, engineers, supervisors, maintenance staff, and office administrators
- Different skill levels and availability that affect task assignment and execution

3. Machines and Equipment

- Industrial machines, production lines, and automated tools
- Machine conditions such as temperature, vibration, and operational status

4. Resources

- Energy, raw materials, inventory levels, and workforce hours
- Resource availability and usage constraints

5. System Constraints and Conditions

- Time limitations, cost constraints, and operational policies
- Unexpected events such as machine failures or task delays

A – Actuators

Actuators are the mechanisms that enable the intelligent agent to interact with the environment and implement decisions. The main actuators in the AI Task Automation System include:

1. Task Assignment Actuator

- Assigns tasks to suitable workers or machines
- Updates task states (pending → in progress → completed)

2. Workflow Automation Actuator

- Automatically triggers dependent tasks
- Manages approvals and workflow transitions

3. Maintenance Actuator

- Schedules preventive maintenance
- Sends maintenance alerts
- Temporarily halts machines when necessary

4. Resource Management Actuator

- Adjusts manpower allocation
- Optimizes energy and material usage

5. Notification Actuator

- Sends alerts and reports to supervisors
- Confirms normal system operations

S – Sensors

Sensors enable the AI agent to perceive the environment and gather real-time information required for intelligent decision-making. Key sensors include:

1. Task Status Sensors

- Monitor task progress, priority, and dependencies

2. Machine Condition Sensors

- Measure temperature, vibration, and operational health

3. Resource Usage Sensors

- Track energy consumption, material usage, and workforce availability

4. Workflow Monitoring Sensors

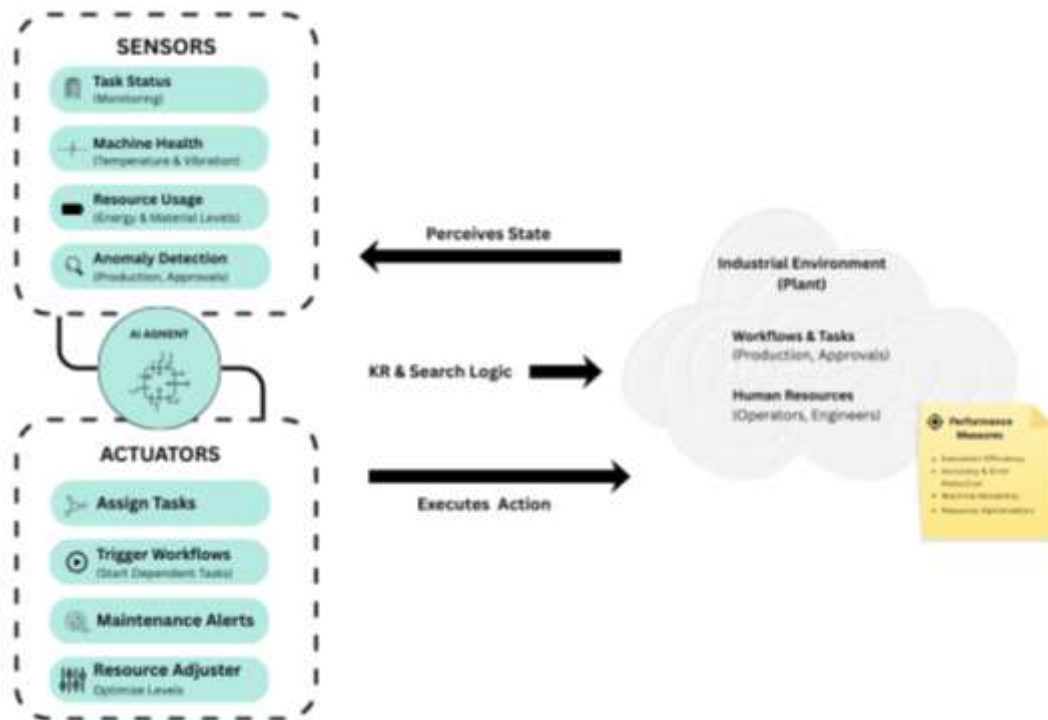
- Detect delays, bottlenecks, and process completion

5. Anomaly Detection Sensors

- Identify abnormal patterns and system faults

RELATIONSHIP DIAGRAM FOR PEAS

Figure bellow illustrates the PEAS model representation for the AI Task Automation System. Sensors continuously collect operational data from the industrial environment, such as task status, machine conditions, resource usage, and data for anomaly detection. This data is processed by the AI agent, which evaluates performance objectives and selects appropriate actions. Actuators then execute these decisions by assigning tasks, triggering workflows, optimizing resources, or issuing alerts. The environment responds to these actions, and updated conditions are again captured by sensors, creating a continuous feedback loop that supports intelligent decision-making and system optimization.



<Figure 1: PEAS Relation Diagram>

REPRESENTSTION OF PEAS IN POC

The Proof of Concept (POC) for the AI Task Automation System illustrates the functioning of the intelligent agent within a straightforward yet realistic industrial environment. The prototype emphasizes the simulation of task automation, oversight, and decision-making by utilizing the PEAS (Performance measure, Environment, Actuators, Sensors) framework. Each aspect of PEAS is embodied through distinct functional elements in the POC, guaranteeing consistency with the system's goals and design.

Performance Measure (P) Representation:

In the proof of concept, performance metrics are illustrated through visual signals and system logs that assess the efficiency of the AI agent in automating tasks and enhancing operations.

The prototype features a fundamental performance dashboard that monitors:

- Duration of task completion (pre- and post-automation)
- Count of tasks successfully executed
- Identified errors or irregularities
- Efficiency of resource utilization (simulated)

For instance, the POC logs execution time and completion status when the system automatically assigns tasks or initiates workflows. A comparison view shows how automation minimizes manual involvement and delays. Despite being simplified, these measurements undeniably demonstrate increases in accuracy, efficiency, and dependability, confirming the system's ability conceptually.

Environment (E) Representation:

The POC environment depicts a simulated industrial workspace featuring workflows, human roles, machines, and resources.

In the model:

- Digital representations exist for tasks such as production jobs, approvals, and maintenance requests.
- Human resources (operators, engineers, managers) are depicted as user roles in the system interface.
- Machine statuses (active, inactive, error) and resource quantities (power, supplies) are emulated through established parameters.

The POC environment enables interaction between various pieces, demonstrating how changes in one component (i.e, a machine problem or delayed approval) affect the total operation. This controlled simulation accurately represents real-world industrial circumstances while remaining acceptable for demonstration and testing purposes.

Actuators (A) Representation:

Actuators within the POC are depicted by system actions that the AI logic triggers automatically.

The prototype showcases essential actuator functions, including:

- Automatic allocation of tasks to suitable users or devices.
- Initiating workflows, where subsequent tasks commence upon the completion of prerequisites.
- Alerts for maintenance are produced when unusual conditions are identified.
- Suggestions for resource modification based on workload or availability

For example, when a task's status changes to "completed," the POC automatically initiates the following task in the process. Similarly, if a simulated machine abnormality is identified, the system will send a maintenance warning. These activities highlight how the AI agent affects the environment by making decisions without human intervention.

Sensors (S) Representation:

Sensors in the POC function as data tracking modules that consistently gather and refresh system data.

The model features artificial sensors for:

- Monitoring task status (pending, in progress, finished)
- Indicators of machine health (normal or abnormal condition)
- Utilization rates of resources
- Hiccups and hold-ups in workflow

These sensors provide live information to the system, enabling the AI logic to understand the present condition of the environment. For instance, a postponed task or unusual machine status is promptly identified and shown, allowing the system to react via suitable actuators. This illustrates the perceptual ability of the intelligent agent

In the Proof of Concept, the PEAS attributes are illustrated via a cohesive simulation of monitoring, decision-making, and action implementation. The **Sensors** identify the state of tasks, tools, and assets, while the system processes data utilizing Knowledge Representation and Search Logic. Moreover, **Actuators** carry out roles such as task allocation and workflow initiation, while **Performance Metrics** evaluate the effectiveness of these actions. This directly reflects the connections shown in the diagram and showcases how each PEAS feature is implemented within the prototype.

CONCLUSION

In conclusion, the PEAS model provides a clear and structured way to define how the AI agent operates within the AI Task Automation System. The performance measures guide the agent toward efficiency, accuracy, and reliability, while the environment represents the dynamic industrial setting where tasks, machines, and resources constantly change. Through sensors, the agent continuously perceives real-time data such as task status, machine conditions, resource usage, and anomalies. Using this information, the agent reasons and decides on the best actions, which are executed through actuators such as task assignment, workflow triggering, maintenance alerts, and resource optimization.

Overall, the AI agent behaves as a goal-oriented, intelligent decision-maker that continuously senses, decides, and acts to automate tasks, reduce errors, improve coordination, and maintain optimal operations. By representing the solution precisely using the PEAS model, the system demonstrates how AI can effectively manage complex industrial processes in a structured, adaptive, and efficient manner.

PEER REVIEW

Peer Review on Team Work	
Reviewer: Tahiya Nuzhath Khan	
Team Members	Category
Malika	4
Aria	4
Pashmia	4